Rishi

650-761-3864 rishivisweswar.boppana@sjsu.edu rishi-boppana rishiboppana

Education

San Jose State University **Jan 2025 – May 2027**
*Master of Science in **Applied Data Science** GPA: 3.75* *San Jose, CA, USA*

- **Coursework**: Generative AI, Data Warehousing, Data Analysis, Deep Learning, Machine Learning, Statistics, Distributed Systems, Big Data

Mahindra University **Oct 2020 – Aug 2024**
*Bachelor of Technology in **Computer Science*** *Hyderabad, India*

Technical Skills

Languages & Frameworks: Python, SQL, JavaScript, React, Node.js, Express, FastAPI, REST APIs
AI & LLMs: Qwen3, vLLM, RAG, LangGraph, LoRA/QLoRA fine-tuning, multi-agent systems, pgvector
Data & Databases: PostgreSQL, MySQL, Redis, Kafka, Apache Spark, MLlib, Parquet, MLflow, Ray Tune
Cloud & DevOps: AWS (EC2, S3, RDS), Docker, GitHub Actions, Git, JMeter load testing
Practices: JWT authentication, caching, query optimization, performance tuning, distributed systems

Experience

Research Assistant **Jul 2026 – Sep 2026**
SJSU Research Foundation *San Jose, CA*

- Developed an **AI-powered patent title generation system** that turns complex patent abstracts into concise, technically accurate titles aligned with **DWPI** standards.
- Fine-tuned **Qwen, Llama, DeepSeek, and T5** on **6,700 patent records** using **FFT, LoRA, and 4-bit QLoRA**; an automated **LLM-as-a-Judge** workflow on a 100-patent test set showed up to **80%** improvement and preference over reference titles.

Student Research Assistant **Jun 2025 – Aug 2025**
San José State University *San Jose, CA*

- Architected a **5-agent AI video editing system** using **Python, LangGraph, Llama 3.2, LLaVA, Whisper, and FFmpeg**, automating multimodal analysis and highlight generation to reduce manual editing effort by **78%** across **50 videos**.
- Built asynchronous **FastAPI** services with **Celery and Redis** for parallel agent execution, task scheduling, and caching, supporting **20 concurrent jobs** with **2.3× higher throughput**.
- Developed a multimodal ranking pipeline using **CLIP embeddings**, speech transcripts, and LLM relevance scoring, improving highlight-selection **Precision@5** from **0.64 to 0.77** across **200 annotated samples**.
- Deployed containerized services on **AWS (EC2, S3, CloudWatch)** using **Docker, GitHub Actions, pytest**, and OpenAI Codex-assisted development, achieving **97.5%** job success and **34%** lower p95 latency.

Software Engineer – Backend **Aug 2024 – Dec 2024**
Narkha LLP *Bangalore, India*

- Developed and maintained **RESTful APIs** using **Python, FastAPI, PostgreSQL, and SQLAlchemy** for financial transaction tracking and reporting, implementing **JWT authentication**, input validation, and role-based access control.
- Built asynchronous data-processing workflows using **Celery and Redis**, handling **100K+ financial records monthly** with automated retries, transaction deduplication, and error logging to improve processing reliability.
- Optimized **PostgreSQL** queries through indexing, SQL query tuning, and pagination, reducing report generation latency by **30%** and improving API performance for transaction history and financial summaries.
- Supported backend deployment using **Docker and AWS EC2**, implementing unit and integration tests with **pytest**, **GitHub Actions CI/CD**, and application logging to streamline releases and troubleshooting.

Data Science Intern **Jun 2023 – Aug 2023**
Innominds Ltd *Hyderabad, India*

- Redesigned a **Python LiDAR pipeline** for a Qualcomm autonomous-vehicle PoC using **RANSAC** ground-plane segmentation and **DBSCAN** clustering (**Open3D, PCL**), extending detection range **25%** and cutting latency **15%**.
- Improved object detection accuracy by **27%** through point-cloud filtering, feature engineering, and hyperparameter tuning.

Projects

Multi-Agent AI Engineering Tutor | *Qwen3, vLLM, PostgreSQL, pgvector, RAG*

- Built a multi-agent tutor that analyzes private codebases and docs with **hybrid RAG** and personalized learning paths.
- Raised local **Qwen3** throughput from **3.6 to 20.6 tokens/sec** (**5.7x**) with **vLLM** on HP ZGX Spark.
- Added persistent agent memory with **PostgreSQL + pgvector** across web, Chrome, and VS Code.

Kayak Agentic Website | *LangGraph, FastAPI, Redis, Kafka*

- Cut redundant queries **60%** with **Redis** caching on a distributed **FastAPI** backend.
- Used **Kafka** pipelines for async price updates and search analytics.

Large-Scale Recommendation Engine | *Spark, Parquet, MLlib, MLflow, Ray Tune*

- Cut **Spark** ETL runtime **60%** (20 to 8 min) over **25M** ratings via AQE, skew-join optimization, and dynamic partitioning.
- Built a **Bronze/Silver/Gold Parquet** lakehouse and tuned an **ALS** model with **MLflow** to **0.641 RMSE**.

Airbnb Clone | *React, Node.js, Express, MySQL, Docker, AWS*

- Built **REST** APIs with **JWT** and transaction-safe bookings on **Node.js, Express, MySQL**, deployed on **Docker** and **AWS**.
- Sustained **200 req/s** with sub-second latency via caching; held **5,000 concurrent users** in **JMeter** tests.

Research

Master's Thesis: Diagnosing failures in **multi-agent AI systems** and improving **agent coordination** in these systems.

Certifications

Amazon Web Services: Certified Developer – Associate | Certified Machine Learning Engineer – Associate